

Hubble – Next-Gen Campus & College Operating System

PROJECT REPORT & TECHNICAL ARCHITECTURE SPECIFICATION

Live URL:	https://hubble-35ts.vercel.app/
Repository:	shreyalukk/hubble (GitHub)
Status:	Fully Operational & Deployed on Vercel
Date:	July 2026

1. Executive Summary

Hubble is an all-in-one, enterprise-grade Campus Operating System built to eliminate communication fragmentation, empower peer-to-peer student engagement, simplify academic administration, and solve critical real-world challenges faced by higher education institutions.

Combining dual-role authentication (Students vs. Teachers), real-time presence-aware messaging, private hub passcode security, an academic Assignment LMS, an Engineering Reference Book & PYQ Library, a verified Notice Board (featuring Hackathons and Internship Drives), a Department Timetable Portal, and a Campus Marketplace, Hubble bridges the gap between campus administration, faculty, and student life.

2. Problem Statement (Real-World Campus Challenges)

Modern universities and engineering colleges face severe operational inefficiencies caused by fragmented tools:

- **Fragmented Communication & Noise:** Important academic notices get buried in spammy WhatsApp groups or unread email threads. Offline/unverified users receiving messages leads to delays.
- **Scattered Study Materials & PYQs:** Course notes and Previous Year Questions (PYQs) are scattered across short-lived Google Drive links, lost between semesters.
- **Inadequate Assignment Tracking:** Students frequently miss homework deadlines across multiple subjects while teachers lack a unified digital LMS to collect and grade work cleanly.
- **Missed Opportunities (Hackathons & Internships):** Flagship national hackathons (e.g., Christ University's *HackChrist v4.0*), coding contests, and off-campus internship drives fail to reach interested students in time.
- **Lack of Timetable & Room Clarity:** Students struggle to locate daily lecture rooms, lab halls, and end-semester examination schedules across semesters.
- **Wasted Educational Resources:** Students spend thousands on expensive new textbooks and calculators that seniors no longer use, with no trusted platform to buy/sell locally on campus.

3. Technology Stack Architecture

Layer	Technology	Key Capabilities & Purpose
Core Framework	Next.js 16 (App Router)	Full-stack React framework with Server Components for optimal SEO, speed, and serverless API routing.

Language	TypeScript 5.x	End-to-end type safety, preventing runtime errors across client components and server actions.
Styling System	Tailwind CSS	Custom responsive UI design system with modern aesthetic cards, micro-animations, and warm dark/light modes.
Backend & Database	Supabase (PostgreSQL)	Managed PostgreSQL database storing users, hubs, messages, assignments, resources, and notices.
Real-Time Engine	Supabase Realtime	WebSocket presence engine tracking active logged-in users and real-time message deletion broadcast.
Security & Auth	Supabase RLS & Auth	Dual-role auth with strict PostgreSQL policies isolating data between users, students, and faculty.
Hosting & CI/CD	Vercel Platform	Automated global edge deployment, instant git-push CI/CD pipelines, and zero-downtime serverless hosting.

4. Key Functional Modules

■ Dual-Role Authentication & Onboarding

Captures department, year of study (1st - 4th), and learning interests for students, while capturing teaching experience and subjects taught for teachers.

■ Real-Time Presence & Message Guarding

Restricts direct messaging strictly to active logged-in users with live online badges (■ Logged In). Features real-time DELETE policies for user message removal.

■ Private Hub Security & Custom Passcodes

Auto-generates join codes (e.g. HUB-8F3A29) or custom creator passcodes. Includes an interactive Passcode Join Modal on the Groups dashboard.

■ Campus Assignments LMS (/assignments)

Allows teachers to post homework with due dates and max points. Students submit file/Drive links, track submission status (Pending/Submitted/Graded), and view countdowns.

■ Engineering Book & PYQ Library (/resources)

Filterable by engineering department (CSE, ISE, ECE, EEE, ME, CIVIL, AI & DS). Features pre-populated reference textbooks (Silberschatz, Horowitz & Sahni, Sedra & Smith, B.S. Grewal) and PYQ papers.

■ Verified Notice Board & Circulars (/announcements)

Pre-loaded with flagship event circulars: Christ University HackChrist v4.0 (■1,50,000 prize pool), National Coding Championship, and Software Engineering Internship drives (■45,000/month stipend).

■ Department Class & Exam Timetable Portal (/timetable)

Displays weekly lecture schedules with time slots, classroom/lab hall numbers (CS-301, Lab 4), and faculty names alongside End-Semester examination schedules.

■■ Campus Peer-to-Peer Marketplace (/marketplace)

Peer-to-peer campus marketplace for buying, selling, or trading used textbooks, lab coats, calculators, and tech gear locally on campus.

5. Database Schema & Relational Tables

- **users:** Stores id, full_name, avatar_url, role ('student', 'teacher', 'admin'), department, college_id.
- **hubs & hub_members:** Stores group metadata, privacy mode ('public', 'private', 'passcode_protected'), passcode, and member roles.
- **direct_messages:** Stores direct chat records with sender_id, receiver_id, content, and timestamp.
- **assignments & submissions:** Stores assignment details, deadlines, student file submissions, and points.
- **resources:** Stores reference books, PYQs, file URLs, and associated engineering department tags.
- **announcements:** Stores official notices, categories ('hackathons & contests', 'internships & hiring', 'exams & timetables'), and pinning status.